

EE 5239 Introduction to Nonlinear Optimization Syllabus

Mingyi Hong

Fall 2024

Lecture Time and Location

Monday and Wednesday, 01:00 PM - 02:15 PM

Course website: See [Canvas](#)

Course location: Keller Hall 3-115

Instructor Contact

[Mingyi Hong](#),

Keller Hall 6-109,

Office Phone: 612-625-3505,

Email: [mhong at umn.edu](mailto:mhong@umn.edu),

Office hours: Mondays 2:15pm – 3:00pm, or by appointment

[Jiaxiang Li](#),

Keller Hall 5-151,

Email: [li003755 at umn.edu](mailto:li003755@umn.edu),

Office hours: Wednesdays 4:00 pm – 5:00 pm, or by appointment

TA Contact

Bingqing Song

Email: [song0409 at umn.edu](mailto:song0409@umn.edu)

Office hours: **Tue and Thur 11: 00am – noon**

Office hour location: Keller 2-276

Textbook

1. Dimitri Bertsekas, "[Nonlinear Programming](#)" (**either V2 or V3 works**)
2. Jorge Nocedal and Stephen Wright, "Numerical Optimization"(Reference)
3. Leon Bottou, Frank E. Curtis, Jorge Nocedal, "Optimization Methods for Large-Scale Machine Learning" (Reference)

4. Additional materials (papers, lecture notes, etc) will be provided throughout the course

Prerequisites

Math 2373, Math 2374, EE 3025 and Matlab (or C, Python)

Required Background

Familiar with basic concepts in linear algebra, mathematics analysis, optimization and statistical modeling; Comfortable in using one of the scientific programming language/environment: MATLAB, R, C++, Python, etc.

Course Objectives

1. To present the basic theory and modern algorithms, concentrating on results that are useful in computation;
2. To give students the background required to use the methods in their own research or engineering work;
3. To give students a thorough understanding of how optimization problems are solved, and some experience in solving them.
4. To introduce modern applications of optimization, including decentralized optimization and information processing, sparse/lower-rank optimization, and optimization applications in (deep) machine learning

Course Detail

- Unconstrained Optimization
 - Optimality Conditions
 - Gradient/Newton's Method
 - Conjugate Gradient Method
 - Stochastic Gradient Descent
 - Momentum Based Methods
 - Convexity, and related concepts
 - Application of supervised learning (linear regression)
 - Application of unsupervised learning (clustering)
- Constrained Optimization
 - Optimality Condition
 - Descent Methods

- Conditional Gradient Method
 - Block Coordinate Descent Method
- Lagrangian Multiplier Theory
 - Equality Constrained Problems
 - Inequality Constrained Problems
 - Sensitivity Analysis
 - Linearly Constrained Problems
 - Duality Theory
- Lagrangian Multiplier Method
 - Penalty Method
 - Method of Multipliers
 - Alternating Direction Method of Multipliers
 - Issues of Large Scale Optimization
- Accelerated, Incremental Proximal Gradient (PG) Method, Subgradient Method, Stochastic Optimization
- Decentralized and distributed optimization
 - Algorithms based on consensus
 - Algorithms based on primal-dual methods
- Optimization issues in machine learning, and neural networks

Home works and Mid Terms

- Tentative Mid Term Slot: 10/30
- 6 - 7 assignments

Learning Outcomes

After completing the course, the students will learn the following

- To appreciate the role of optimization in real-life decision making process
- To understand the basic and advanced properties of nonlinear problems
- To understand the basic and advanced analysis of nonlinear optimization methods
- To solve a mathematical model by various modern optimization methods
- To pick the best way to analyze and solve a given optimization problem

Grading

- Numerical grade = Homework (30%) + exam (35%) + class project (30%) + bonus points (up to 5%)
- **Two** homeworks have to be properly **typed**, and should be **submitted electronically**, in pdf format.
- After homework or exams are graded, an announcement will be posted on Canvas. You will have **one week** to discuss with the TA or me about your grades. After one week, all grades are finalized.

Email Correspondence

Some of you may prefer to send the instructor or the TAs emails for short questions. You should understand that although we will respond to all the emails we receive, it may not be possible for us to give you a reply within 24 hours. So you are encouraged to talk to us during office hours to get immediate feedback

Collaboration Policy

- In-class and after-class discussions are strongly encouraged.
- One homework submission per person. Copying of others' homework is not allowed.
- No collaborations during the exams.

Academic Honesty Statement

The ECE Department has an expectation that all students will be honest in their actions and communications. Individuals suspected of committing academic dishonesty will be directed to appropriate university offices as per University policy. For more information regarding Academic Misconduct see <https://communitystandards.umn.edu/know-code/disciplinary-process>

Professionalism Statement

The ECE Department has an expectation that all students will behave in a professional manner during all interactions with fellow students, faculty, and staff. Treating others with respect and having constructive communications are examples of being professional

Disabilities Resources

If you have a documented disability and need special accommodations, feel free to contact the instructor or the student disability resources at <https://diversity.umn.edu/disability/home>